A hybrid feedforward-feedback controller for SMA bending wire

1. The introduction

This work focuses on the modeling, simulation, and control of a Shape Memory Alloy (SMA) bending wire actuator. Shape Memory Alloy materials exhibit significant hysteresis loops, and standard negative-gain Proportional-Integral (PI) controllers can introduce severe actuation delays (up to 5.0 seconds) when thermal dynamics and convective cooling are ignored.

To overcome this latency, feedforward control profiles use two 1-D lookup tables extracted via First-Order Reversal Curve (FORC) measurements to capture the core non-linear phase physics of the system. While this approach minimizes response lag, it introduces two distinct operational vulnerabilities:

1. **Idealized Instantaneous Jumps:** Relying purely on direct temperature command signals forces non-physical, instantaneous temperature jumps whenever the reference trajectory changes.
2. **Load Sensitivity:** 1-D lookup tables are structurally constrained to a single, static load baseline, making the controller susceptible to tracking errors under varying external mechanical forces.

Integrating a physical Joule heating model addresses real-world thermal boundaries, demonstrating how variations in the cooling medium (such as transitioning from ambient air to cooling liquids) affect response rates. Under liquid-convection conditions, the system exhibits a cumulative 0.7-second delay, which includes a distinct 0.3 to 0.4-second dead-band zone. However, implementing safe operational voltages while balancing convective heat transfer introduces tracking errors.

To resolve both the varying mechanical load problem and the tracking errors, expanding the model to a 2-D lookup table framework that incorporates load variations is structurally inefficient. This method exponentially increases the experimental sample size from a baseline of 50 tests to (50 * n) tests (where (n) represents each discrete load variation), incurring prohibitive computational and experimental costs.

Consequently, a hybrid feedforward-feedback control strategy is utilized. This architecture implements a negative-gain feedback PI loop as an auxiliary compensator to handle real-time tracking deviations, leaving the computationally efficient 1-D lookup tables as the dominant driving force behind the actuator plant.

2. The hybrid feedforward-feedback control system

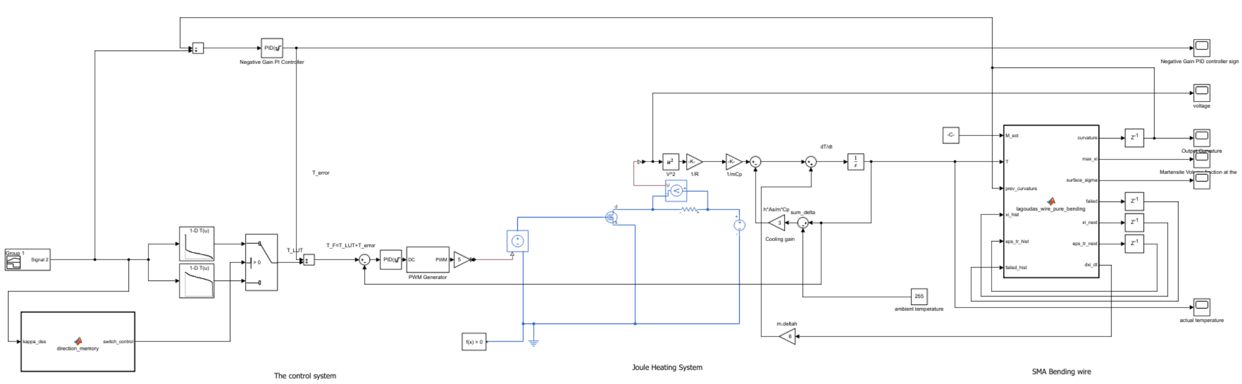

Figure (1) The complete control loop of the system

The overall control loop consists of three primary components: the control subsystem, the Joule heating setup, and the SMA bending wire plant.

Subsystem Architecture

- **The Control Subsystem:** This block combines a parallel feedforward-feedback architecture. The 1-D lookup tables serve as the dominant feedforward driving force to handle the material's nonlinearities. Operating alongside them, an auxiliary negative-gain PI controller acts as a feedback mechanism to eliminate remaining transient deviations.
- **The Joule Heating Subsystem:** This component receives the reference temperature signal generated by the hybrid controller to regulate the electrical power delivery required to drive the wire toward the targeted curvature profile.
- **The SMA Bending Wire Plant:** This block models the actual physical actuator, which undergoes thermal contraction or yields to the applied external mechanical loads depending on its internal temperature.

Operating Boundaries and Baseline

The baseline for this experimental configuration is established using the 1-D lookup tables, which were captured at a static load sufficient to bend the tip of the wire to a curvature of 52 m^{-1}. Any operational tracking deviations below this 52 m^{-1} baseline threshold are dynamically compensated for by the auxiliary PI loop. To maintain system stability and prevent integrator windup, the auxiliary PI controller output is constrained to a saturation limit of ±30 K and incorporates an anti-windup clamping method.

3. Robustness Testing

To evaluate the operational boundaries and robustness of the control configuration, systemic testing is conducted across different load profiles. Initial characterization establishes system behavior at the baseline load of 52 m^{-1}. Subsequent experiments systematically increase and decrease the applied mechanical load to determine tracking performance under varying stress states.

Case (a) the load at 52 m^{-1}load baseline

This case evaluates the baseline performance of the hybrid feedforward-feedback architecture under the reference load condition .The test observes the dynamic adjustments introduced by combining the dominant feedforward lookup tables with the auxiliary PI controller, which is implemented specifically to eliminate residual steady-state tracking errors

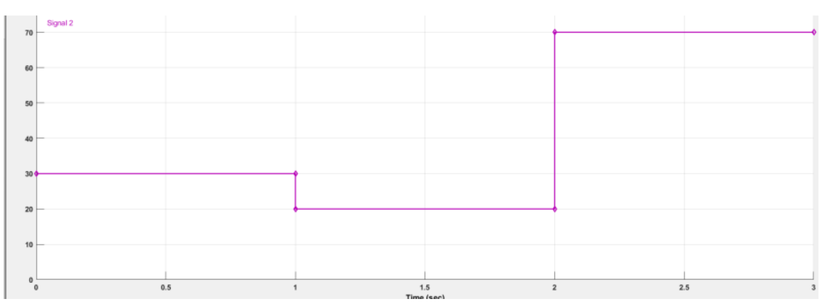

Figure (2) The input desired trajectory

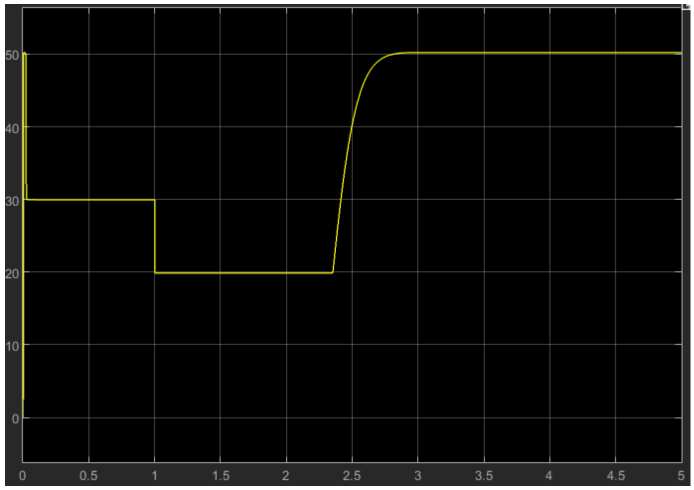

Figure (3) The output curvature

The profile initializes with the wire in a pre-bent condition of 52 m^{-1} induced by the baseline mechanical load. Upon activation, the system successfully executes a step transition down to a 30 m^{-1} curvature and subsequently tracks down to a 20 m^{-1} step. However, during the final command phase, the actuator is unable to exceed its structural boundaries to achieve the targeted 70 m^{-1} curvature step due to the physical operational limits of the wire assembly.

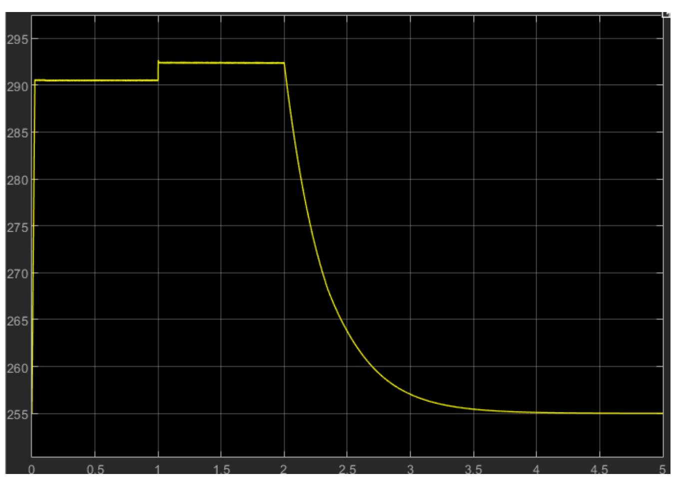

Figure (4) the temperature curve

A distinct thermal spike occurs during the 20 m^{-1} step transition. This profile confirms that the auxiliary PI loop temporarily assumes a primary driving role to accelerate the system toward its target destination.

- **Sustained holds:** The controller rapidly cuts down power input once target curvatures are achieved.
- **Phase stability:** The minor thermal drop is insufficient to trigger an un-bending phase transformation.
- **Saturation boundaries:** During the 70 m^{-1} command, the auxiliary loop operates at maximum capacity.
- **Mechanical blocks:** The system remains structurally limited at a baseline of 52

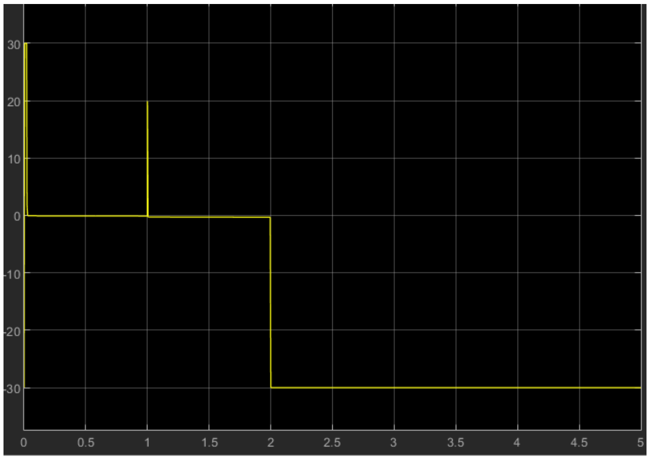

Figure (5) The negative gain auxiliary PID controller output

The auxiliary loop output details the source of the thermal spikes. The controller delivers a maximum command signal to the Joule heating system to accelerate transitions, rapidly dropping to a lower value once the targeted curvature stabilizes.

Upon receiving the 70 m^{-1} step command, the feedback controller immediately saturates at maximum capacity. It maintains this peak output for the remainder of the simulation window. Because the mechanical load dictates a structural boundary of 52 m^{-1}, the wire cannot deflect further, leaving a persistent tracking error that keeps the controller saturated.

Case (b) decreasing the load under the 52 m^{-1} load baseline

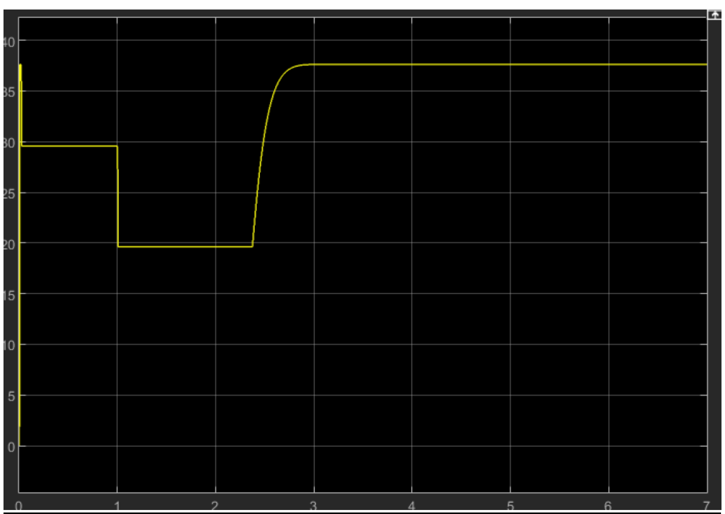

Figure (6) the output curvature with decreased load

Under this reduced loading profile, the initial mechanical deflection bends the wire to approximately 37 m^{-1}. Upon activation, the actuator tracks down through the 30 m^{-1} step and continues to the 20 m^{-1} step. It then hits its physical limit while transitioning toward the 70 m^{-1} target. A minor tracking deviation of roughly 1 m^{-1} occurs during both the 20 m^{-1} and 30 m^{-1} steady-state holds.

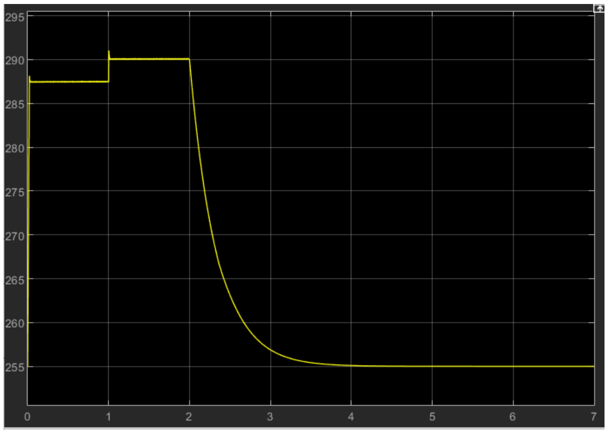

Figure (7) the temperature curve

The experimental data confirms the presence of stress-dependent transformation temperatures under reduced loading conditions. Because the lower mechanical stress decreases the thermal energy required to reach a specific curvature target, the auxiliary PI controller automatically scales down the temperature reference command to prevent overheating and maintain optimal tracking.

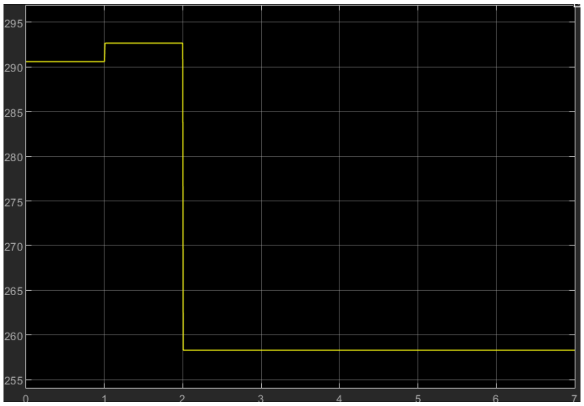

Figure (8) The lookup tables output

A comparative analysis of the two profiles reveals the clear influence of the auxiliary feedback loop in modulating the reference input. The controller dynamically updates the command signal to the Joule heating system, adjusting for instantaneous tracking errors to ensure the wire accurately follows the target trajectory

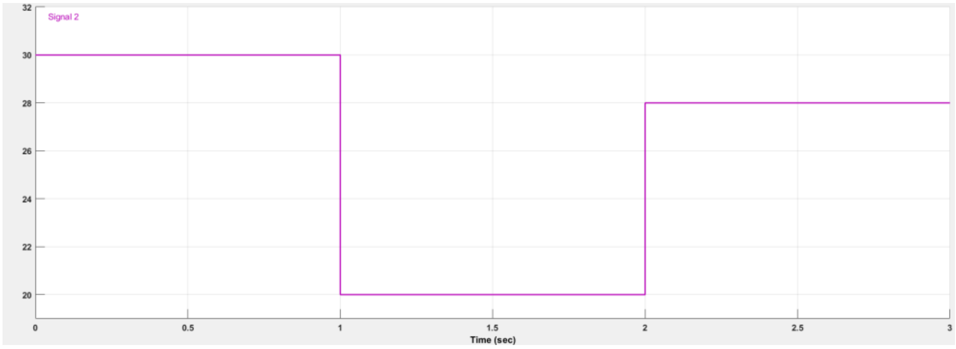

Figure (9) The inbounds trajectory in decreased load case

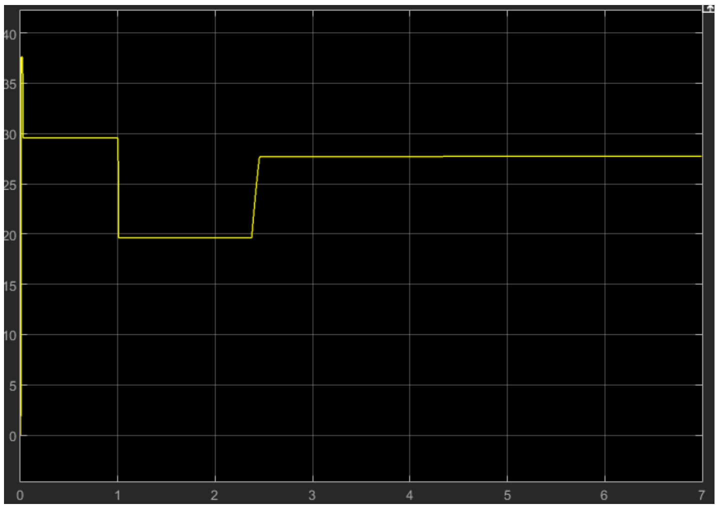

Figure (10) the output curvature of the inbounds desired trajectory

The system exhibits highly precise trajectory tracking, maintaining a small, consistent steady-state error of less than 1 m^{-1} throughout the controlled steps. While further refinement of the PI controller gains could potentially minimize this residual deviation, the current tracking accuracy falls well within acceptable engineering tolerances for this actuator application.

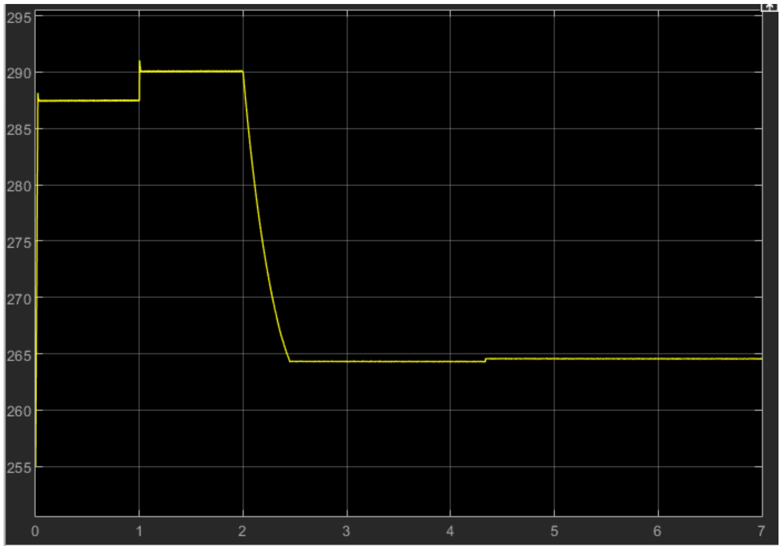

Figure (11) The temperature curve

The profile displays a minor temperature increase toward the end of the simulation window. However, this slight thermal rise does not negatively impact system performance, as the magnitude remains well below the threshold required to trigger an unintended material phase transformation from martensite to austenite.

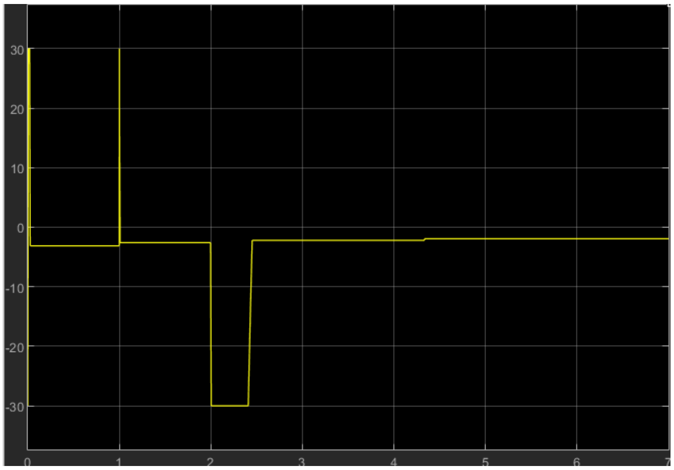

Figure (12) The negative gain PI controller output

The control signal profiles reveal the origin of the minor temperature rise observed at the end of the simulation. Because this transient adjustment remains small enough to preserve the intended material phase state, it has no impact on overall system tracking performance.

4. Safety operating and the balancing between the electric power and the cooling method

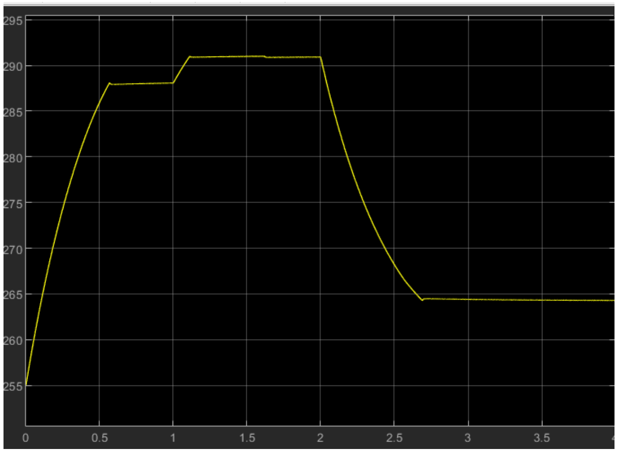

Figure (13) The temperature output with safety consider at 3 V operating

The thermal response profile demonstrates a significantly slower temperature rise during the initial step, resulting in a 0.5-second tracking lag. Additionally, the system successfully eliminates the non-physical, instantaneous temperature jumps during subsequent step transitions. To optimize this behavior in practical deployment, the wire should be pre-heated to a baseline condition immediately below its austenite start temperature (As), keeping the actuator primed for rapid movement regardless of active reference commands.

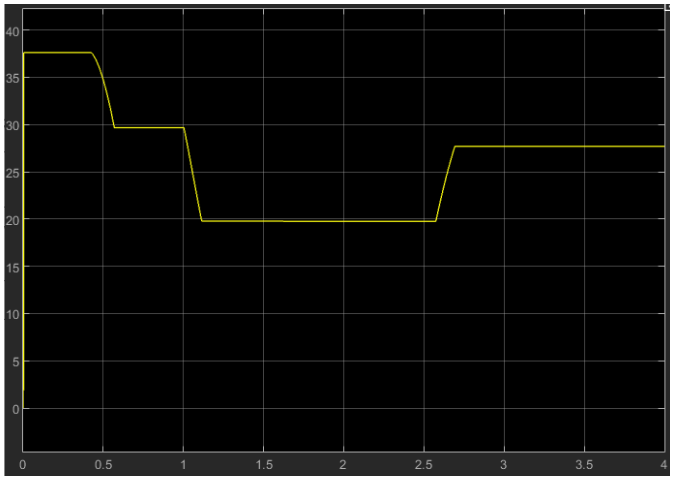

Figure (14) the output curvature

The system achieves smooth trajectory tracking and steady-state stability via the auxiliary PI loop without requiring complex adjustments to the inner-loop temperature controller. This control approach successfully addresses the low-voltage tracking deviations that arose when the operating potential was lowered to satisfy safety requirements. The results demonstrate that the hybrid feedforward-feedback architecture reliably compensates for these constraints, making it highly effective for real-world deployments where both safe voltages and aggressive cooling methods are applied.

5. The conclusion

The primary objective of this three-part documentation series was to develop a practical, highly applicable control methodology for a Shape Memory Alloy (SMA) bending wire actuator, focusing on predicting and mitigating the physical anomalies encountered in real-world deployment

Thermal and Power Balancing

The investigation demonstrated that convective heat transfer serves as a primary governing factor in real-world environments. Environmental convection introduces significant response delays, particularly during passive cooling steps. To manage this behavior, the system must tightly balance convective cooling gains against electrical power inputs to ensure optimal performance without exceeding material thresholds.

Initial testing deliberately utilized high electrical power inputs to isolate and analyze convective cooling dynamics independently. Once these thermal boundaries were established, the system parameters were balanced to maximize trajectory tracking efficiency while completely safeguarding the wire against localized overheating and permanent crystalline degradation.

Control Strategy and Plant Robustness

Following plant characterization, a hybrid feedforward-feedback control architecture was implemented. This configuration delegates primary tracking commands to two 1-D lookup tables extracted via the First-Order Reversal Curve (FORC) measurement method. These tables serve as the dominant driving force, handling the severe inherent nonlinearities of the material. Operating in parallel, an auxiliary negative-gain PI controller functions as a compensator, dynamically suppressing transient tracking errors and correcting real-time deviations.

Framework and Implementation Style

To bridge the gap between theoretical model-based design and practical application, this series evaluated complex thermal and material nonlinearities through an isolated, modular framework. Every systemic component—including the core SMA bending plant, the electro-thermal Joule heating equations, and the environmental convection boundaries—was tested and verified as a standalone case.

This individual validation process culminated in evaluating system performance under varying external mechanical forces. The final experimental stages confirmed that the auxiliary PI controller reliably compensates for fluctuating bending loads, providing a robust foundation for successful transition from simulation to real-world hardware implementation.